

Attack based on Operational Error caused by Stop-Signal Reaction Time in VR space

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ABSTRACT

This addresses emerging threats arising from the widespread adoption of Virtual Reality (VR) technology, focusing on attacks targeting VR controllers that exploit cognitive factors, such as the human reaction speed, through malicious programs. In particular, it examines attack methods designed to induce misoperations by considering users' reaction time limitations. The objective of this study is to analyze malicious actions in VR spaces caused by misoperations related to human reaction time and to elucidate the impacts and risks of such threats targeting VR technology.

Index Terms: Human Reaction Time, Misoperation Induction, Stop-Signal Reaction Time.

1 INTRODUCTION

One psychological tactic for inducing user misoperations, known as "dark patterns" [3, 5], involves deceptive user interface designs that trick users into performing unintended actions. Countermeasures against such issues typically include improving user information literacy and implementing security measures at the platform level. However, focusing solely on the psychological aspects of attacks, such as dark patterns, is insufficient. Therefore, we investigated the feasibility of attacks that target human cognitive factors, particularly the limitations of reaction speed.

With the increasing adoption of Virtual Reality (VR) technology, the issue of misoperations in VR environments has gained significant attention.[1] In particular, there is a growing possibility of attacks targeting VR controllers that exploit human cognitive limitations, such as the limitations of reaction speed. These attacks considers users' reaction speed limitations to induce misoperations or unintended actions. Moreover, because VR interactions often involve larger and more dynamic physical movements compared to smartphones or computers, the impact of such attacks could be severe in VR environments. Unintended actions may cause risks, such as system misuse or information leakage.

This study focuses specifically on reaction times, particularly the Stop-Signal Reaction Time (SSRT)[2, 4], which represents the delay between perceiving a sensory stimulus and halting an intended action. Attackers can strategically time the insertion of malicious advertisements or similar stimuli to bypass user intentions and trigger unintended actions. To successfully execute such attacks, it is necessary to either "guide" the user's tapping timing or "predict" it using sensors or comparable technologies.

In this research, we developed a tool to simulate and analyze these threats, enabling us to assess the feasibility of attacks that exploit SSRT.

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2 MISOPERATION-INDUCING ATTACK CAUSED BY STOP-SIGNAL REACTION TIME

2.1 Motivation

This study specifically examines attack methods that exploit the limitations of user reaction speeds to induce misoperations. These methods utilize human cognitive limitations, particularly delayed reaction times, to trigger unintended actions, which may result in risks, such as system misuse or information leakage. Because VR interactions often involve larger and more dynamic user movements, including those performed with VR controllers or hand tracking, attackers can easily exploit reaction time delays to induce misoperations. Consequently, the impact of such attacks is more severe in VR environments compared to smartphones and computers.

This paper investigates unauthorized actions in VR spaces caused by misoperations related to human reaction times. The objective is to elucidate the impacts and risks of threats that exploit misoperations targeting VR technology.

2.2 Stop-Signal Reaction Time (SSRT)

The Stop-Signal Task (Fig. 1) is a method used to evaluate executive functions and inhibitory control, with a focus on sustained attention and behavioral inhibition. This task involves requiring participants to stop an ongoing action when a specific signal is presented. In the task, participants are required to perform an action, such as pressing a button when a stimulus, such as a lamp lighting up, is displayed (Go task). Conversely, when a different stimulus, including a beep sound, occurs, they inhibit the action of pressing the button (Stop task).

Humans have the ability to inhibit actions, enabling them to cancel an action they are about to execute. However, there is a delay in successfully halting the action, and depending on the timing, the inhibition may fail. This phenomenon, which reflects information processing in the brain, is explained by the race model proposed by Logan and Cowan [2]. The race model describes how the brain processes this information: one process attempts to execute the action (Go process), while another process attempts to interrupt it (Stop process). If the Stop process is completed before the Go process, the action is successfully stopped. However, if the Go process is completed first, the action continues as planned.

The time required to stop the action is referred to as the Stop-Signal Reaction Time (SSRT). The SSRT can be estimated using the Stop-Signal Task. According to Logan and Cowan's research, the average SSRT is approximately 200 milliseconds.

2.3 Our Attacks Caused by SSRT in VR Space

We investigate the implementation of attacks in VR spaces that exploit operational errors caused by human reaction times. As part of a comprehensive analysis of attack methods, we classified the scenarios in which such attacks could occur into the following three patterns.

- (A) Insertion of misoperation objects during repeated actions, such as rapid button-pressing. (guidance)

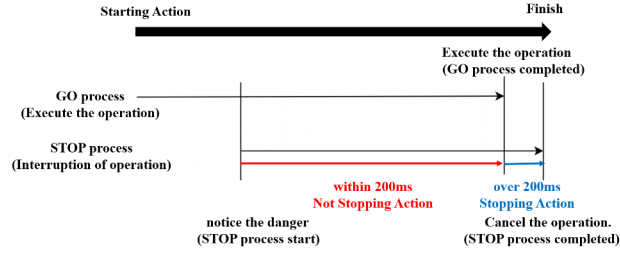


Figure 1: Description of Stop-Signal Task and Stop-Signal Reaction Time (SSRT)

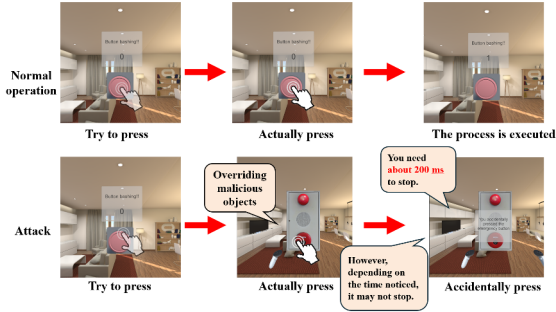


Figure 2: Attack on Rapid Button-Bashing Game

- (B) Insertion of misoperation objects at the moment a button is about to be pressed during a timing-based game. (guidance)
- (C) Detection of button-press behavior using sensors, followed by insertion of misoperation objects. (prediction)

In this paper, we develop experimental environments for “guidance-based” tap timing, specifically focusing on (A) rapid button-pressing games and (B) timing-based games.

2.3.1 Attack on Rapid Button-Bashing Game

In a rapid button-bashing game, to induce erroneous operations, the system introduces decoy objects (objects that players should avoid pressing) during the process of repeatedly pressing buttons to increase a numerical value. The occurrence of an error is determined by whether the decoy object has been pressed.

This approach aims to simulate malicious attacks (e.g., redirecting users to harmful websites), but under controlled experimental conditions; in particular, these behaviors are reproduced to avoid actual harm.

The system measures the distance between hand (or controllers) and the designated activation range of buttons or misoperation objects. If a hand or controller is detected as having “touched” the button or misoperation object, it either increments the button press count or triggers an interaction with the misoperation object.

Fig. 2 illustrates the overall configuration of this behavior, showing the phenomenon where users press a misoperation object during rapid button-bashing. In particular, cognitive factors, such as attention dispersion caused by rapid pressing and recognition errors in identifying the intended target are included as mechanisms that caused erroneous operations.

2.3.2 Attack on Timing-Based Game

In a timing-based game, to induce misoperations through timing, the timing-based game includes a reset button, allowing users to



Figure 3: Attack on Timing-Based Game

retry the game repeatedly. The method for detecting button taps is similar to that used in the rapid button-bashing game.

As shown in Fig. 3, users are instructed to press a button with precise 5-second timing. Between 4.8 seconds and 200 milliseconds before the target time, a misoperation object appears with a certain probability.

While the rapid button-bashing game induces misoperations by leveraging unconscious reactions during fast operations, the timing-based game differs by targeting user attention immediately before actions that require concentration and precision.

The probability of incorrect operation objects appearing is set low, and a reset button enables repeated trials, minimizing users’ awareness of these incorrect operation targets. Specifically, the probability of incorrect operation objects appearing is set at 1/3.

2.3.3 Attack By Predictive Misoperation Induction

Although not implemented in this study, we propose a predictive method for inducing misoperations: the hand-tracking function of Meta Quest 3 enables continuous monitoring of user movements. Therefore, we suggest a method for displaying a misoperation object when the user’s movement exceeds a specific acceleration threshold. This method is relatively easy to implement, as it does not rely on machine learning techniques.

3 COUNTERMEASURE

The erroneous operation induction methods demonstrated in this study exploit human cognitive limitations, making it difficult for users to consciously implement countermeasures. Consequently, it is essential for platforms to adopt appropriate measures to effectively mitigate these risks.

For example, in situations where users might unintentionally click on an advertisement, implementing a mechanism that temporarily disables interactions immediately after the ad is displayed can help reduce such risks.

Furthermore, when requesting permissions required for an application, granting these permissions should involve more than a single tap. Combining a tap with additional actions can prevent unintentional permission grants. Alternatively, even when permissions are granted through a single action, mechanisms, such as requiring a long press, can provide users with an opportunity to reconsider their decision, allowing them to cancel the action. Such mechanisms serve as effective safeguards.

4 CONCLUSION

In this study, we introduced a new threat linked to the widespread adoption of VR technology, focusing on attacks that target the operation of VR controllers and exploit cognitive factors, particularly the limitations of human reaction speed, to induce misoperations via malicious programs.

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